# 4 Chemical potentials and quantum statistical physics – bosons and fermions

The problems are roughly in order of difficulty. Those indicated with ♡ are ones which are highly recommended. Any labelled with ♣ are the hardest ones, which might only occur as a "sting in the tail" at the end of a long examination question.

♡*Problem 4.1* **Chemical potentials and the Saha equation**
The notion that two systems which exchange particles must have equal chemical potentials in equilibrium may be used in discussing, as a function of temperature, the degree of ionisation of a gas of atomic hydrogen or the degree of dissociation of molecules into their constituent atoms.

Consider the case of the ionisation of a hydrogen gas. We will regard the density or temperature to be such that the bound hydrogen and the unbound protons and electrons behave as classical gases. We will make the assumption that we can neglect the excited states of bound hydrogen, so the atom is either in its ground state or ionised.

(i) Regarding the bound hydrogen gas as having chemical potential $\mu_{\text{H}}$ and the ionised gas as having chemical potential $\mu_{\text{I}}$, what relation must they obey in equilibrium?

We will regard the "Hydrogen universe" as electrically neutral. Consider the free energies

$$F_0(T, V, N_{\text{H}}, N_{\text{I}}) \quad \text{and} \quad F(T, V, N_{\text{H}}, N_{\text{p}}, N_{\text{e}}) \ ,$$

where $N_{\text{H}}$, $N_{\text{I}}$, $N_{\text{p}}$ and $N_{\text{e}}$ are the numbers of H atoms, ionised H atoms, protons and electrons in the universe.

(ii) Write down the expression for $\mathrm{d}F$ and deduce an expression for the chemical potentials, $\mu_{\text{H}}$, $\mu_{\text{p}}$ and $\mu_{\text{e}}$ in terms of partial derivatives of $F$. Assuming neutrality so that $\mathrm{d}N_{\text{p}} = \mathrm{d}N_{\text{e}} = \mathrm{d}N_{\text{I}}$, perform the same analysis with $\mathrm{d}F_0$ and its partial derivatives and deduce a relation between $\mu_{\text{p}}$ and $\mu_{\text{e}}$, and $\mu_{\text{I}}$.

Now consider the simplified Hamiltonian for the H atom

$$\mathcal{H} = \frac{p^2}{2m_{\text{H}}} - \varepsilon_0 \ ,$$

where $m_{\text{H}}$ is the mass of the H atom and $\varepsilon_0$ is the binding energy.

(iii) Write down the partition function for a (bound) H atom and hence its free energy.

(iv) Write down the free energy for a classical ideal gas composed of $N_{\text{H}}$ such H atoms (remembering indistinguishability) and hence deduce the chemical potential for an H atom.

(v) Regarding the atoms, protons and electrons as ideal classical gases of identical particles use the result of the previous part of this question and the relationship between the chemical potentials derived in part (ii) to show

$$\lambda_{\mathrm{Te}}^3 \lambda_{\mathrm{Tp}}^3 \, n_{\mathsf{p}} n_{\mathsf{e}} = \lambda_{\mathrm{TH}}^3 \, n_{\mathrm{H}} \mathrm{e}^{-\beta\varepsilon_0} \ ,$$

where for particle species $j$

$$n_j = \frac{N_j}{V} \quad \text{and} \quad \lambda_{\mathrm{T}j} = \left( \frac{\mathsf{h}^2}{2\pi m_j k_{\mathrm{B}} T} \right)^{1/2} \ .$$

(vi) Recognising the neutrality of the gas and the disparity in mass between the electrons and protons show (a form of Saha's equation)

$$n - n_{\mathrm{I}} = \mathrm{e}^{\beta\varepsilon_0} \lambda_{\mathrm{Te}}^3 n_{\mathrm{I}}^2 \ ,$$

where $n = N/V$ and $N$ is the total number of protons – free or bound in H atoms.

(vii) Hence show that

$$n_{\mathrm{I}} = \frac{-1 \pm \sqrt{1 + 4\mathrm{e}^{\beta\varepsilon_0} \lambda_{\mathrm{Te}}^3 n}}{2\mathrm{e}^{\beta\varepsilon_0} \lambda_{\mathrm{Te}}^3} \ ,$$

choosing the appropriate sign.

(viii) Deduce the expressions for $n_{\mathrm{I}}$ as $T \to 0$ and $T \to \infty$. Comment on the behaviour of $n_{\mathrm{I}}(T, V, N)$ in the limit that $V \to \infty$ with $N$ and $T$ fixed.

*Problem 4.2* **The Yang-Yang equation**
Show from the First law, including variable particle number, that

$$C_{V,N} = T \left( \frac{\partial S}{\partial T} \right)_{V,N} \ .$$

From the Gibbs-Duhem relation,

$$-S\mathrm{d}T - N\mathrm{d}\mu + V\mathrm{d}p = 0$$

establish the *Yang-Yang* equation

$$C_{V,N} = -NT \left( \frac{\partial^2 \mu}{\partial T^2} \right)_{V,N} + VT \left( \frac{\partial^2 p}{\partial T^2} \right)_{V,N} \ .$$

♡*Problem 4.3* **Neutrinos**
Neutrinos are, for practical purposes, massless non-interacting spin $1/2$ fermions with energy $E_{\mathbf{p}} = \mathrm{c}|\mathbf{p}|$ and they are non conserved. A peculiarity is that only *one* of the spin states exists in Nature, because of the breakdown of parity in the weak interaction.

(i) Calculate the density of single particle states of the system and express the internal energy, $U$, of the gas of neutrinos in terms of an integral over the density of states and the Fermi occupancy factor.

(ii) Using the fact that

$$\int_0^\infty \mathrm{d}x \, \frac{x^3}{\mathrm{e}^x + 1} = \frac{7\pi^4}{120} \ ,$$

show that the internal energy of the system is

$$U = V \frac{7\pi^2}{240} \frac{(k_{\mathrm{B}}T)^4}{\hbar^3 \mathrm{c}^3} \ .$$

(iii) Use this to find the pressure of the neutrino gas.

♡*Problem 4.4* **Ripplons**

The capillary waves on the (two dimensional) surface of a quantum liquid (e.g $^4$He) are excitations which have a dispersion relation of the form $\varepsilon_k = Dk^{3/2}$ and behave like spinless bosons whose number is not conserved.

(i) Explain why the chemical potential, $\mu$, of these excitations must have the value $\mu = 0$.

(ii) Write down an expression giving the mean number $\langle n_k \rangle$ of such bosons present in each single particle state at a temperature $T$ and show that the thermally averaged total number, $\langle N \rangle$ , of such excitations present at a temperature $T$ is given by

$$\langle N \rangle = (k_{\mathrm{B}}T)^{4/3} \left( \frac{A}{3\pi D^{4/3}} \right) \ I$$

where $I$ is the dimensionless integral given by

$$I = \int_0^\infty \mathrm{d}z \, \frac{z^{1/3}}{\mathrm{e}^z - 1}$$

(and which you need not evaluate).

(iii) Show also that the internal energy, $U$, of the system at a temperature $T$ is given by

$$U = (k_{\mathrm{B}}T)^{7/3} \left( \frac{A}{3\pi D^{4/3}} \right) \ J$$

where $J$ is the dimensionless integral given by

$$J = \int_0^\infty \mathrm{d}z \, \frac{z^{4/3}}{\mathrm{e}^z - 1} \ ,$$

(and which you need not evaluate).

(iv) Find also the heat capacity, $C_A$, of the system at constant area.

(v) What does the third law of thermodynamics tell us about the entropy and the heat capacity of a system at low temperatures? Explain how knowledge of the heat capacity of the system allows us to calculate its entropy and hence use the expression you have obtained for $C_A$, to calculate the entropy of the system of excitations.

(vi) Show that the Helmholtz free energy, $F$, and the internal energy, $U$, of this system of capillary waves are related by

$$F = -\frac{3}{4} U \ .$$

*Problem 4.5* **Entropy and fermion occupation numbers**
The fermion occupation number of state $\mathbf{k}\sigma$, $n_{\mathbf{k}\sigma}$, is

$$n_{\mathbf{k}\sigma} = \frac{1}{\mathrm{e}^{\beta(\varepsilon_{\mathbf{k}\sigma}-\mu)}+1} \ . \tag{4.1}$$

(i) Express $\mathrm{e}^{\beta(\varepsilon_{\mathbf{k}\sigma}-\mu)}$ in terms of $n_{\mathbf{k}\sigma}$.

(ii) Express $1+\mathrm{e}^{-\beta(\varepsilon_{\mathbf{k}\sigma}-\mu)}$ (noting the minus sign in the exponent) in terms of $n_{\mathbf{k}\sigma}$.

The Grand energy, $\Omega$, for fermions is

$$\Omega = -k_{\mathrm{B}}T\sum_{\mathbf{k}\sigma}\ln\left(1+\mathrm{e}^{-\beta(\varepsilon_{\mathbf{k}\sigma}-\mu)}\right) \ . \tag{4.2}$$

(iii) Show that

$$\begin{aligned} S &= -\left(\frac{\partial\Omega}{\partial T}\right)_{V,\mu} \\ &= -k_{\mathrm{B}}\sum_{\mathbf{k}\sigma}\left(n_{\mathbf{k}\sigma}\ln n_{\mathbf{k}\sigma}+(1-n_{\mathbf{k}\sigma})\ln(1-n_{\mathbf{k}\sigma})\right) \ , \end{aligned} \tag{4.3}$$

using the identities established in the the first two parts of this problem.

(iv) Using $\Omega = U - TS - \mu N$, determine $S$ in terms of $n_{\mathbf{k}\sigma}$ and show consistency with the last part of the question.

(v) Take differentials of both sides of Eq. (4.3) and deduce that if all occupation numbers are kept constant $S$ does not change.

*Problem 4.6* **Pauli magnetic susceptibility**
Consider a set of spin-1/2 fermions residing in three dimensions experiencing a uniform external field, $H\hat{\mathbf{z}}$, at zero temperature.

The fermions have a magnetic moment

$$\mu_{\mathrm{B}} = \frac{e\hbar}{2m} \ ,$$

(the *Bohr magneton*), absorbing the "1/2" so that the magnitude of the $z$-component of spin is $\sigma = \pm 1$. Then the energy is

$$\varepsilon_{\mathbf{k}\sigma} = \frac{\hbar^2k^2}{2m} - \mu_0\mu_{\mathrm{B}}\sigma H \ ,$$

where we have made the factor of $\mu_0$ explicit.

There is a fixed total number of fermions

$$N = N^{\uparrow} + N^{\downarrow} \ .$$

We assume there is no effect on the motion due to the magnetic field. (In reality there is also the Landau diamagnetism which we shall neglect.)

The question is about the *ground state* of the fermi gas, where there is a *common* fermi energy, i.e. $\varepsilon_{\mathrm{F}}^{\uparrow}(H) = \varepsilon_{\mathrm{F}}^{\downarrow}(H)$, but the number of up and down spin electrons, $N^{\uparrow}(H)$ and $N^{\downarrow}(H)$ are chosen to minimise the total energy.

(i) Sketch the densities of states for the two spin states, $g^{\uparrow}(\varepsilon)$ and $g^{\downarrow}(\varepsilon)$.

(ii) Find the numbers of up- and down-spin electrons, $N^{\uparrow}(H) = N + \Delta(H)$ and $N^{\downarrow}(H) = N - \Delta(H)$, as a function of the external field, *assuming that* $\mu_0 \mu_{\rm B} H \ll \varepsilon_{\rm F}(H = 0)$.

(iii) Hence deduce the magnetisation, in the limit $\mu_0 \mu_{\rm B} H \ll \varepsilon_{\rm F}(H = 0)$,

$$M(H) = \mu_{\rm B} \left( N^{\uparrow} - N^{\downarrow} \right) \ .$$

(iv) Deduce the *Pauli susceptibility*, $\chi = \partial M / \partial H$.

(v) Turning to the case where $\mu_0 \mu_{\rm B} H \not\ll \varepsilon_{\rm F}(H = 0)$, determine the lowest field, $H_{\rm pol}(N)$, at which only the up-spin state is occupied – so the gas is *spin-polarised* and the magnetisation is *saturated*.

*Problem 4.7* **Temperature dependence of $U$, $S$, $F$ and $\mu$ at low temperatures**
Assume that only a shell of fermions of thickness $\Delta\varepsilon \sim k_{\rm B} T$ around $\varepsilon = \mu$ is disrupted at temperature $T$ from either being occupied or empty. Also assume that $C_V = \gamma T$. Then, using standard expressions for the thermodynamic potentials, deduce the low temperature form of:

(i) The internal energy, $U$.

(ii) The entropy, $S$.

(iii) The Helmholtz free energy, $F$.

(iv) The chemical potential, $\mu$.

*Problem 4.8* **Observability of Hawking radiation**
The application of thermal ideas to black holes was introduced by Bekenstein and Hawking in 1973-74. This work resulted in identifying a temperature associated with the black hole ,

$$k_{\rm B} T_{\rm H} = \frac{\hbar c^3}{8\pi G M} \ ,$$

where $M$ is the mass of the black hole.

(i) The mass of the Sun is $M_{\odot} \simeq 2 \times 10^{30}$ kg. What is $T_{\rm H}$ for a (hypothetical) solar-mass black hole. Why would this be difficult to observe?

(ii) What mass of body would be observable in the light of the answer to the last point? What scale of bodies in the Solar System does that correspond to?

Hawking radiation from a black hole is approximately of the form:

$$\frac{{\rm d}M}{{\rm d}t} c^2 = -4\pi R_{\rm S}^2 \sigma n_{\rm ch} T_{\rm H}^4 \ .$$

I.e. loss of energy via black body radiation from an area of the event horizon, $4\pi R_{\rm S}^2$ at the temperature $T_{\rm H}$. The Schwarzschild radius, $R_{\rm S}$, and the Stefan-Boltzmann constant, $\sigma$, are:

$$R_{\rm S} = \frac{2GM}{c^2} \quad \text{and} \quad \sigma = \frac{\pi^2 k_{\rm B}^4}{60 \hbar^3 c^2} \ .$$

$n_{\text{ch}}$ reflects the number of species of possible radiating "massless" particles[1]. For low temperatures, $k_{\text{B}}T < m_{\text{electron}}\text{c}^2$, only photons and neutrinos (and anti-neutrinos) contribute[2].

(iii) Show that the lifetime, $\tau$, of a black hole with initial mass $M_0$ obeys:

$$\tau = \frac{n_{\text{ch}}}{5120\pi}\,\frac{M_0^3}{m_{\text{P}}^3}\,t_{\text{P}}\ ,$$

where the Planck mass, $m_{\text{P}}$, and time, $t_{\text{P}}$, are defined:

$$m_{\text{P}} = \sqrt{\frac{\hbar c}{G}} \simeq 2.2\times 10^{-8}\,\text{kg} \quad \text{and} \quad t_{\text{P}} = \sqrt{\frac{\hbar G}{\text{c}^5}} \simeq 5.4\times 10^{-44}\,\text{s}\ .$$

If $M_0 = M_\odot$, what order is $\tau$? What is the age of the Universe?

(iv) "Primordial black holes" may have been created at the big bang (or shortly after). What mass would they have to be to be decaying now, nearby to Earth (i.e. not observed now at a large red-shift, for example)[3]?

[1] This is a little complicated – it depends on the whether the particles exist in more than one spin state (neutrinos do not, for example). Also if $k_{\text{B}}T_{\text{H}} \simeq m\text{c}^2$ for some particle there are threshold effects on the expression.

[2] The current bounds on the neutrino mass are around $0 < m_{\text{neutrino}}\text{c}^2 < 0.12\,\text{eV}$.

[3] An interesting recent suggestion (Barco, Oscar, *Monthly notices of the Royal Astronomical Society*, **506**, 806 (2021)) links gamma ray bursts with primordial black holes falling into a large black hole.